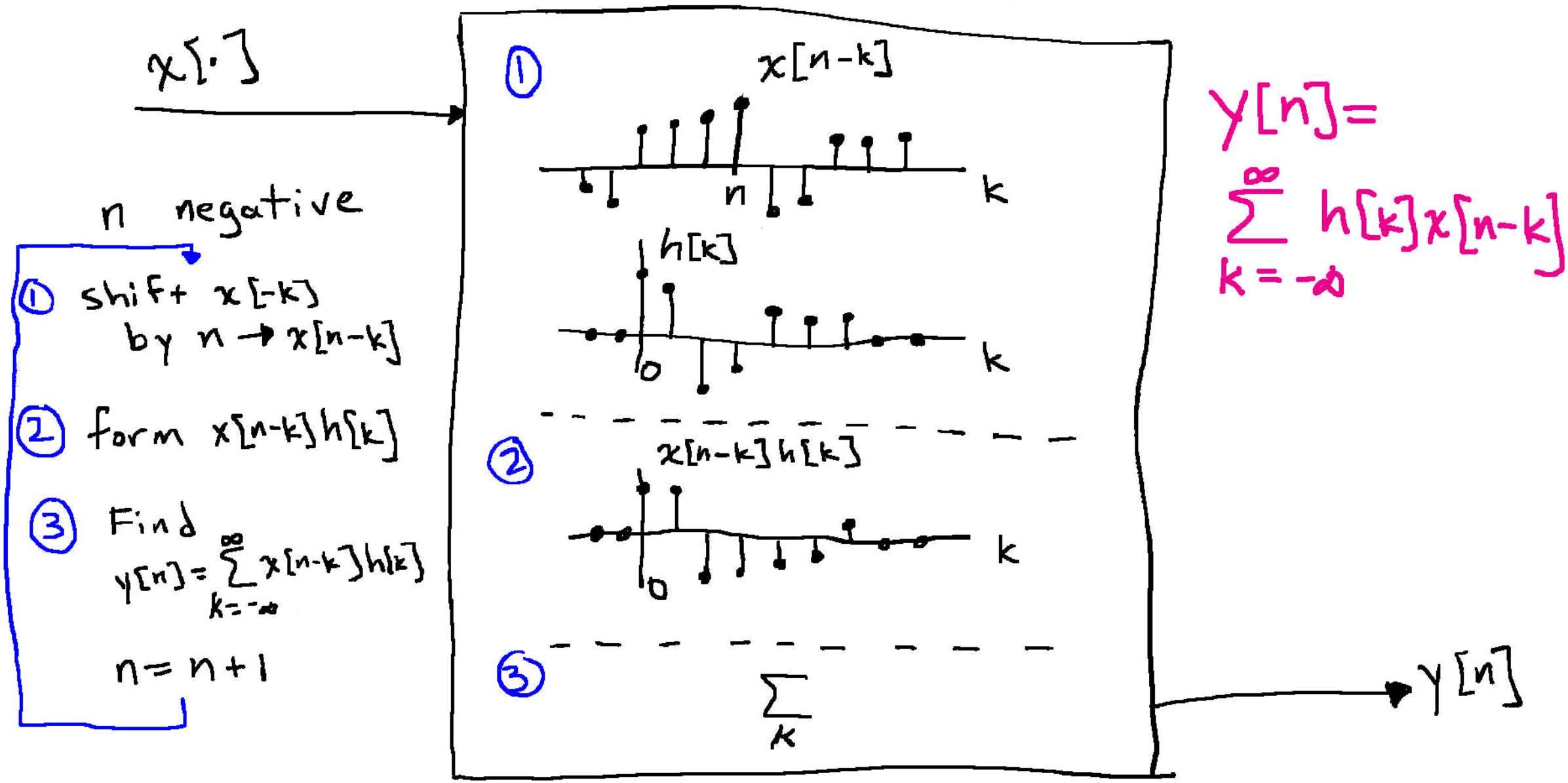
Examples of Graphical Evaluation of Discrete-Time Convolution

Convolution Factory

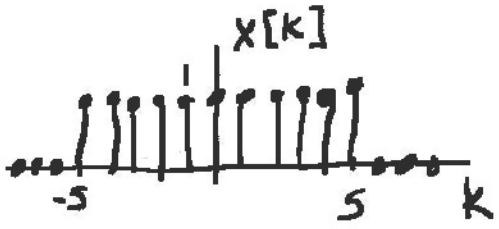


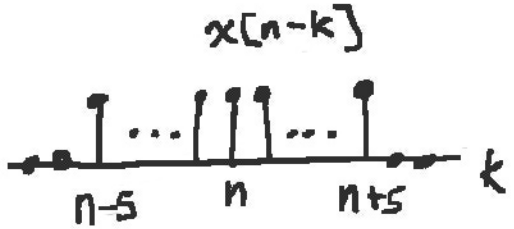
$$y[n] = \sum_{k=-\infty}^{\infty} h[k]x[n-k]$$

Example: Find $y[n] = h[n] * x[n]$

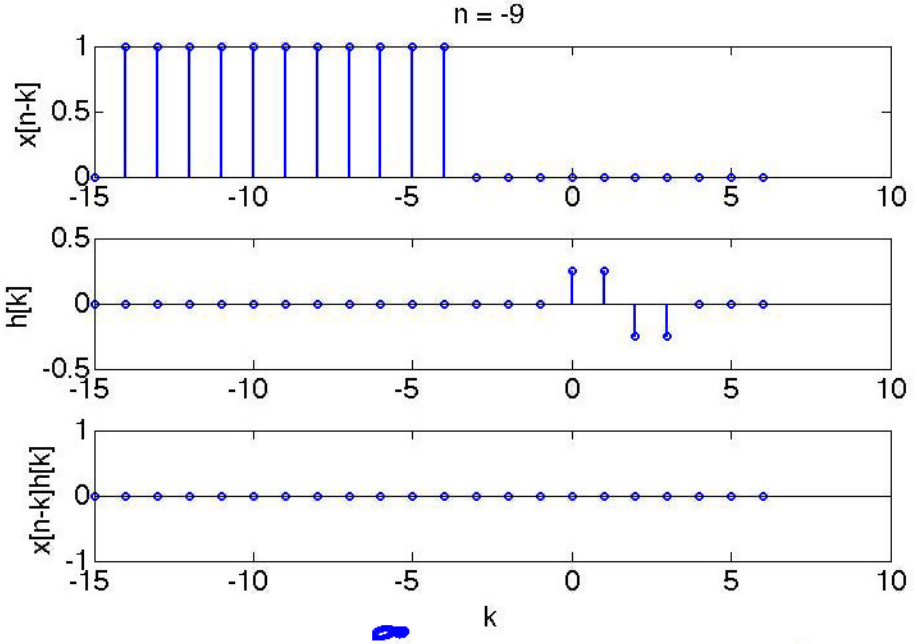
$$x[n] = u[n+5] - u[n-6]$$

$$h[n] = \begin{cases} 1/4, & n = 0, 1 \\ -1/4, & n = 2, 3 \\ 0, & \text{other} \end{cases}$$





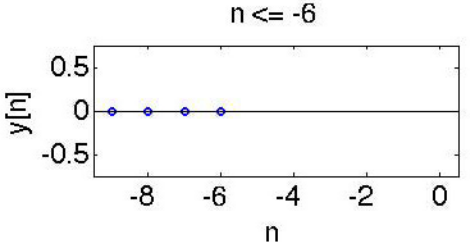
① ② ③



$$y[n] = \sum_{k=-\infty}^{\infty} x[n-k]h[k]$$

$x[n-k]\,h[k] = 0$ for $n \leq 6$

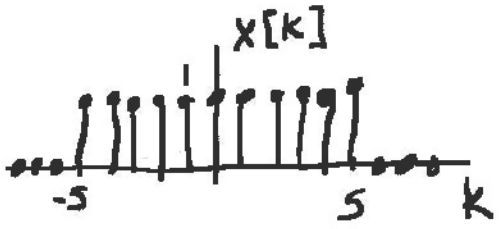
$y[n] = 0$

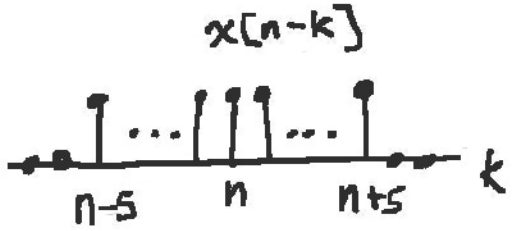


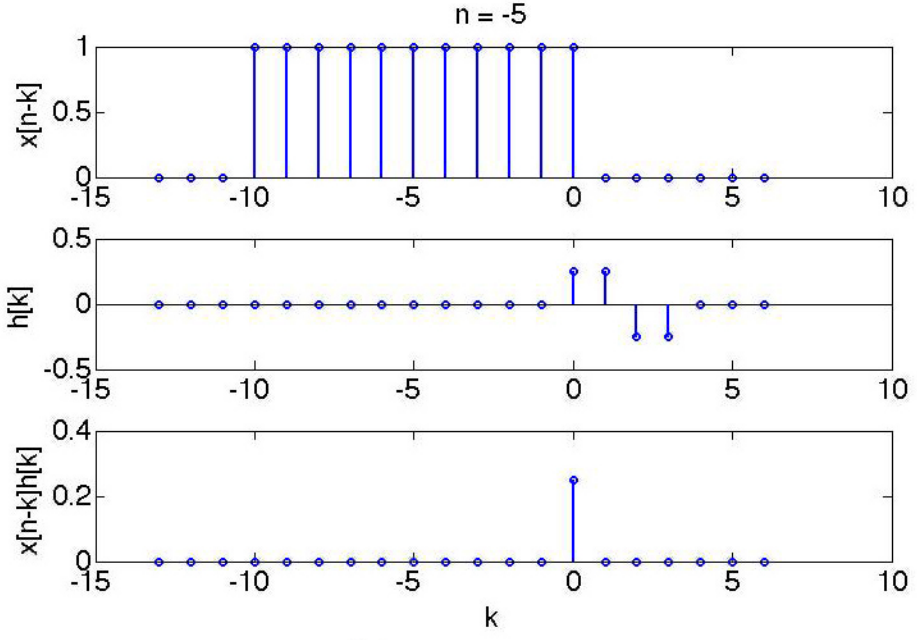
Example: Find $y[n] = h[n] * x[n]$

$$x[n] = u[n+5] - u[n-6]$$

$$h[n] = \begin{cases} 1/4, & n = 0, 1 \\ -1/4, & n = 2, 3 \\ 0, & \text{other} \end{cases}$$

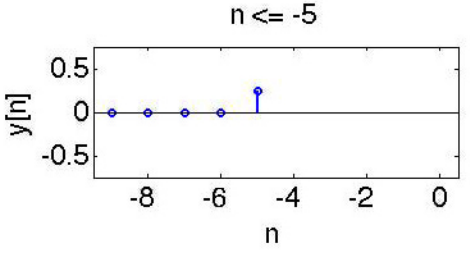




① 

$$x[-5-k]h[k] = \begin{cases} 0 & k \neq 0 \\ 1/4 & k = 0 \end{cases}$$

$$y[-5] = 1/4$$

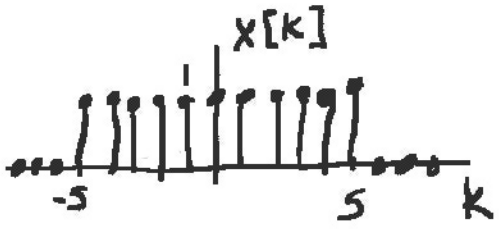
② 

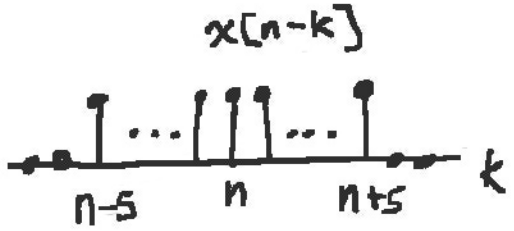
③ $$y[-5] = \sum_{k=-\infty}^{\infty} x[-5-k]h[k]$$

Example: Find $y[n] = h[n] * x[n]$

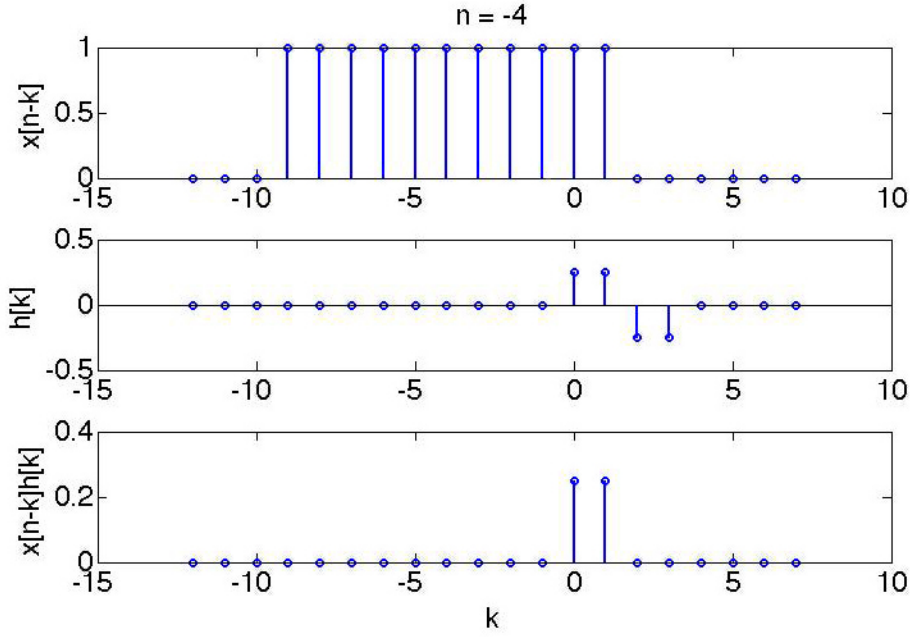
$$x[n] = u[n+5] - u[n-6]$$

$$h[n] = \begin{cases} 1/4, & n = 0, 1 \\ -1/4, & n = 2, 3 \\ 0, & \text{other} \end{cases}$$





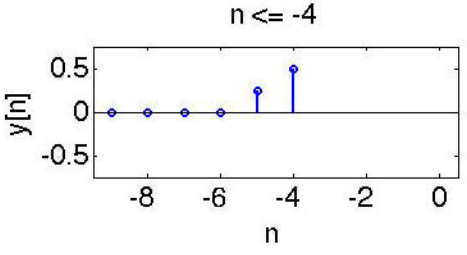
① ②



$$x[-4-k]h[k] = \begin{cases} 0 & k \neq 0, 1 \\ 1/4 & k = 0, 1 \end{cases}$$

$$y[-4] = 1/2$$

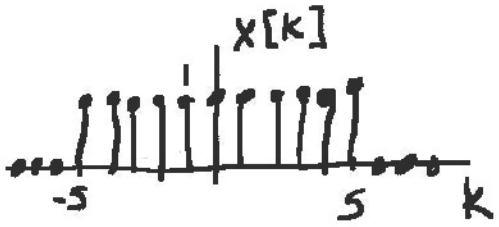
$$y[-4] = \sum_{k=-\infty}^{\infty} x[-4-k]\, h[k]$$

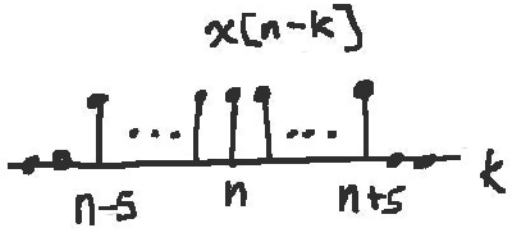


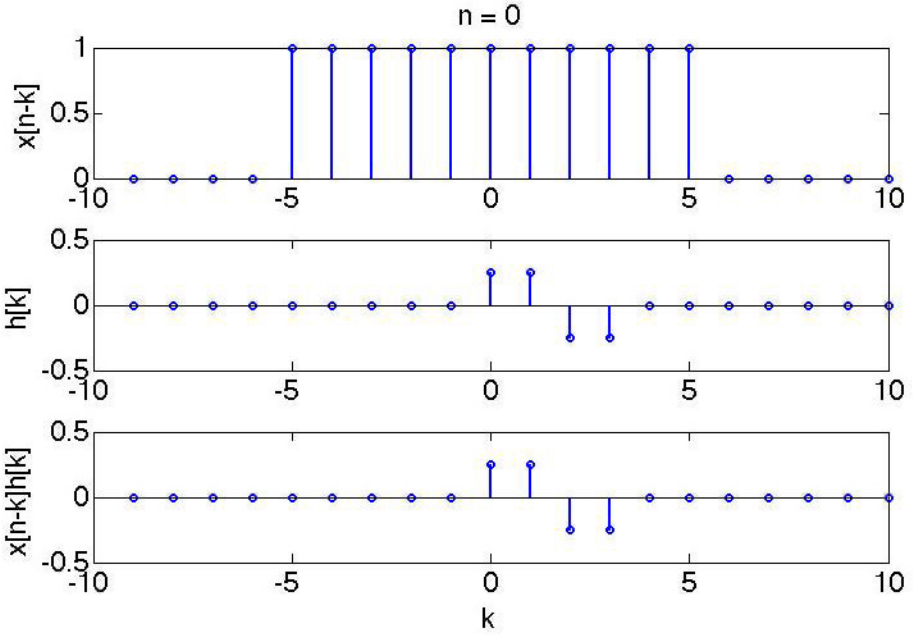
Example: Find $y[n] = h[n] * x[n]$

$x[n] = u[n+5] - u[n-6]$

$$h[n] = \begin{cases} 1/4, & n = 0, 1 \\ -1/4, & n = 2, 3 \\ 0, & \text{other} \end{cases}$$



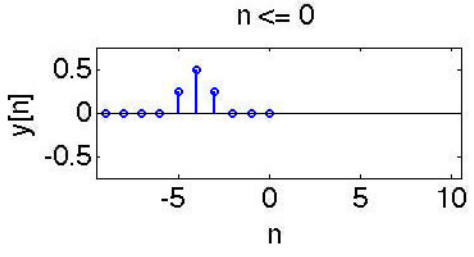


① 

$$x[0-k]h[k] = \begin{cases} 0 & k \neq 0,1,2,3 \\ 1/4 & k = 0,1 \\ -1/4 & k = 2,3 \end{cases}$$

② $y[0] = 0$

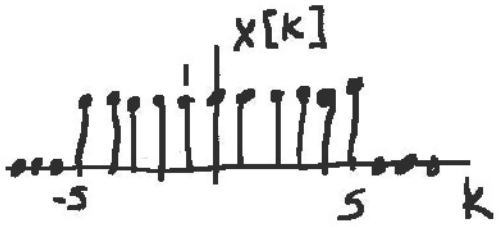
③ $$y[0] = \sum_{k=-\infty}^{\infty} x[0-k]\,h[k]$$

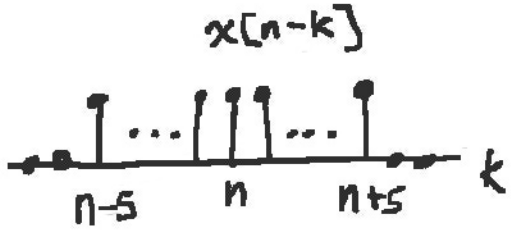


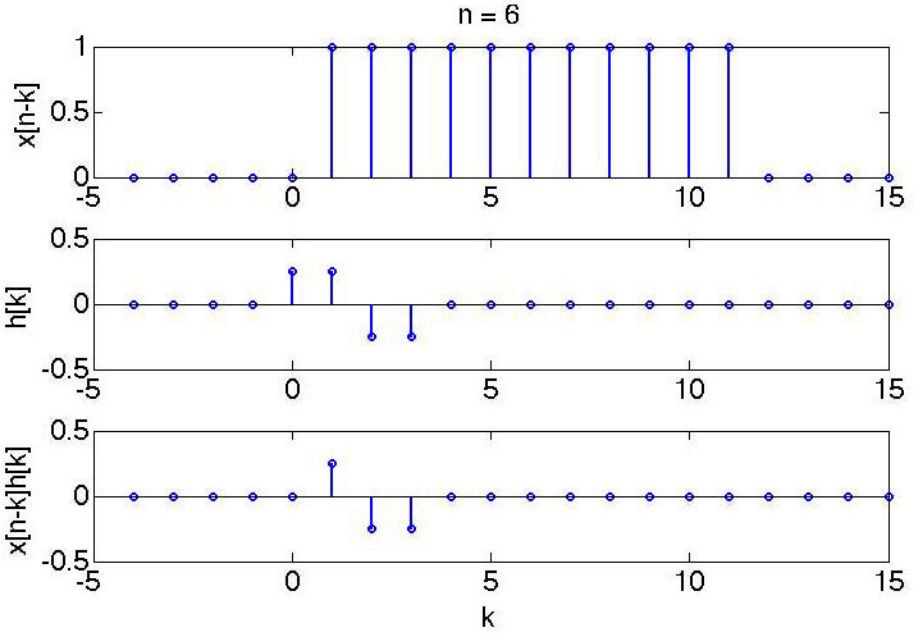
Example: Find $y[n] = h[n] * x[n]$

$$x[n] = u[n+5] - u[n-6]$$

$$h[n] = \begin{cases} 1/4, & n = 0, 1 \\ -1/4, & n = 2, 3 \\ 0, & \text{other} \end{cases}$$

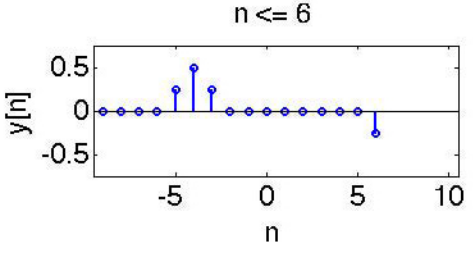




① 

$$x[6-k]h[k] = \begin{cases} 1/4 & k = 1 \\ -1/4 & k = 2, 3 \\ 0 & k \neq 1, 2, 3 \end{cases}$$

$$y[6] = -1/4$$

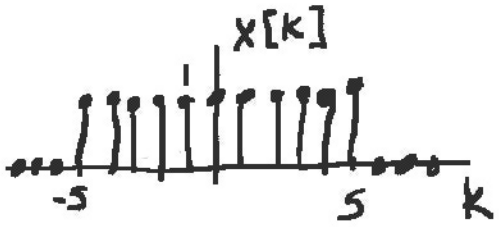
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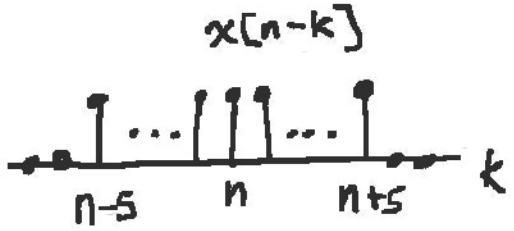
③ $$y[6] = \sum_{k=-\infty}^{\infty} x[6-k]\,h[k]$$

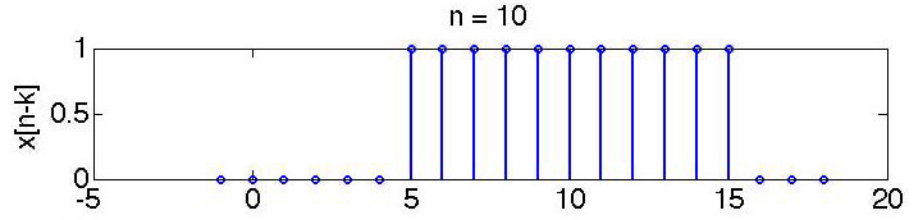
Example: Find $y[n] = h[n] * x[n]$

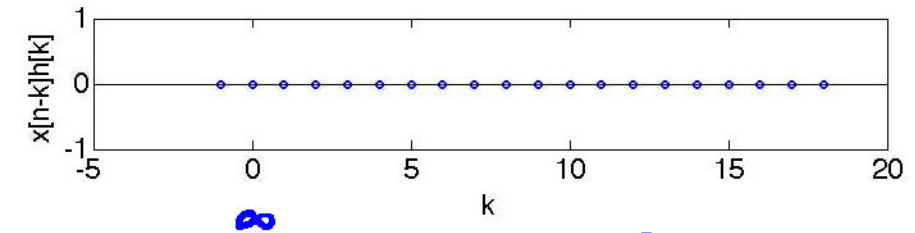
$x[n] = u[n+5] - u[n-6]$

$$h[n] = \begin{cases} 1/4, & n = 0, 1 \\ -1/4, & n = 2, 3 \\ 0, & \text{other} \end{cases}$$





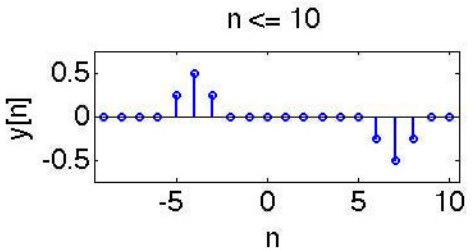
① 

② 

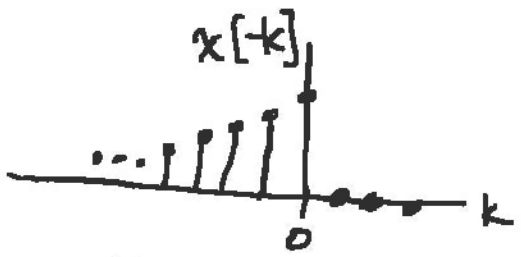
$x[n-k]h[k] = 0$ for $n \geq 9$

$y[n] = 0, \quad n \geq 9$

③ $$y[10] = \sum_{n=-\infty}^{\infty} x[10-k]h[k]$$



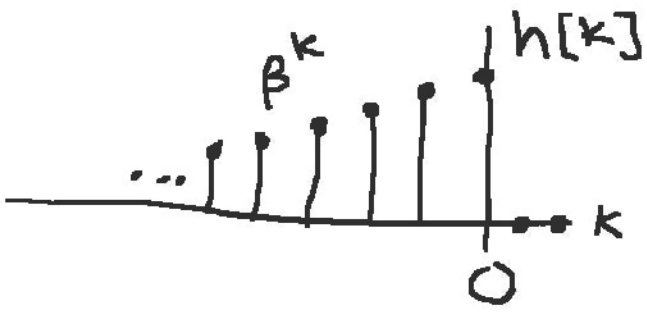
Example: $x[n] = \alpha^n u[n]$, $h[n] = \beta^n u[-n]$ $|\alpha|<1$, $|\beta|>1$



$$x[-k] = \begin{cases} \alpha^{-k}, & k \le 0 \\ 0, & k > 0 \end{cases}$$

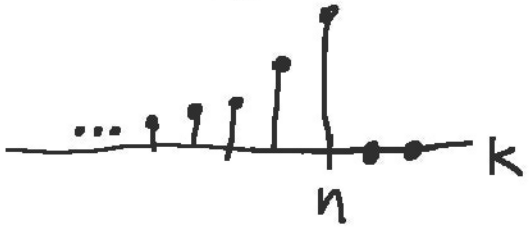


$$x[n-k] = \begin{cases} \alpha^{n-k}, & k \le n \\ 0, & k > n \end{cases}$$



(a) $n<0$

$$x[n-k]h[k] = \begin{cases} \beta^k \alpha^{n-k}, & k \le n \\ 0, & 0 \end{cases}$$



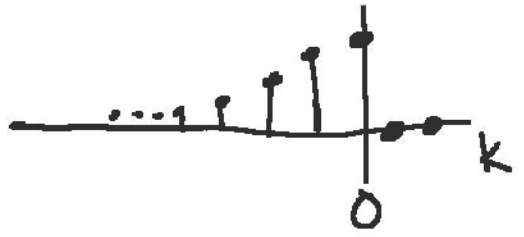
$$y[n] = \sum_{k=-\infty}^{n} \beta^k \alpha^{n-k}$$

let $\ell = -k+n$

$$y[n] = \sum_{\ell=0}^{\infty} \beta^{n-\ell} \alpha^{\ell} = \beta^n \sum_{\ell=0}^{\infty} \left(\frac{\alpha}{\beta}\right)^{\ell} = \beta^n \left(\frac{1}{1-\alpha/\beta}\right)$$

(b) $n \ge 0$

$$x[n-k]h[k] = \begin{cases} \beta^k \alpha^{n-k}, & k \le 0 \\ 0, & k > 0 \end{cases}$$



$$y[n] = \alpha^n \left(\frac{1}{1-\alpha/\beta}\right)$$

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